

Data Assimilation in Nonlinear Fluid Flow III

Nudging: comparable state accuracy with deterministic feedback and far fewer model solves

Presentation 3 of 4

1. Learning objectives

- Derive continuous and discrete nudging updates.
- Interpret nudging as a feedback observer and as a simplified variational correction.
- Prove a basic exponential error-decay result.
- Compare accuracy, error maps, and computational cost with the KF and EnKF baselines.

2. Why look beyond EnKF?

The high-accuracy EnKF baseline achieved mean RMSE 0.012 with $M = 40$ nonlinear forecasts per cycle. For expensive CFD, the forecast cost dominates:

$$C_{\text{EnKF}} \approx M C_{\mathcal{M}}.$$

Can we obtain a useful correction with only one model trajectory?

3. Continuous nudging equation

Start from a forecast model

$$\dot{\mathbf{x}} = \mathcal{F}(\mathbf{x}).$$

Add a feedback term based on the innovation,

$$\dot{\mathbf{x}} = \mathcal{F}(\mathbf{x}) + \mu \mathbf{H}^\top (\mathbf{y} - \mathbf{H}\mathbf{x}).$$

Here $\mu > 0$ is the nudging gain.

Key idea

The model still governs the state; measurements continuously pull it toward observed reality.

4. Discrete nudging at observation times

After one nonlinear forecast,

$$\mathbf{x}_k^f = \mathcal{M}_{\Delta t}(\mathbf{x}_{k-1}^a),$$

apply

$$\mathbf{x}_k^a = \mathbf{x}_k^f + \gamma \mathbf{G}(\mathbf{y}_k - \mathbf{H}\mathbf{x}_k^f).$$

$\mathbf{G}$ spreads observed innovations to the model state.

- If every variable is observed, $\mathbf{G} = \mathbf{I}$ is possible.
- For sparse sensors, $\mathbf{G}$ can interpolate/smooth the innovation spatially.

5. Full-state observation case

If $\mathbf{H} = \mathbf{I}$ and $\mathbf{G} = \mathbf{I}$,

$$\mathbf{x}^a = (1 - \gamma)\mathbf{x}^f + \gamma\mathbf{y}.$$

Thus nudging is a convex blend for $0 \leq \gamma \leq 1$.

- $\gamma = 0$: ignore measurements.
- $\gamma = 1$: replace the forecast with measurements.
- intermediate γ : retain dynamical smoothness while correcting drift.

6. Sparse observations need a reconstruction operator

Our Burgers test observes only 4 of 32 grid points. We form innovations at sensor locations,

$$r_j = y_j - u^f(x_j),$$

and periodically linearly interpolate them to the full grid,

$$\delta u(x) = \mathcal{I}(r_1, \dots, r_m).$$

Then

$$u^a(x) = u^f(x) + \gamma \delta u(x).$$

7. Error dynamics: the key object

Let $\mathbf{x}^*$ be the true state and $\mathbf{x}$ the nudged estimate. Define

$$\mathbf{e} = \mathbf{x} - \mathbf{x}^*.$$

For a linear reference model $\dot{\mathbf{x}} = A\mathbf{x}$ and noiseless observations $\mathbf{y} = H\mathbf{x}^*$,

$$\dot{\mathbf{e}} = (A - \mu H^\top H)\mathbf{e}.$$

The feedback term modifies the error dynamics directly.

8. Theorem: exponential decay under sufficient feedback

Theorem

Assume

$$\frac{A + A^\top}{2} \preceq \alpha I, \quad H^\top H \succeq \eta I$$

on the error subspace of interest, with $\eta > 0$. If $\mu\eta > \alpha$, then

$$\|\mathbf{e}(t)\| \leq e^{-(\mu\eta - \alpha)t} \|\mathbf{e}(0)\|.$$

The observation feedback must dominate the maximum error-growth rate.

9. Proof by an energy estimate

Let

$$V(t) = \frac{1}{2} \|\mathbf{e}(t)\|^2.$$

Then

$$\begin{aligned}\dot{V} &= \mathbf{e}^\top A \mathbf{e} - \mu \mathbf{e}^\top H^\top H \mathbf{e} \\ &\leq \alpha \|\mathbf{e}\|^2 - \mu \eta \|\mathbf{e}\|^2 \\ &= -(\mu \eta - \alpha) \|\mathbf{e}\|^2.\end{aligned}$$

Hence

$$\dot{V} \leq -2(\mu \eta - \alpha) V.$$

Grönwall's inequality gives the stated exponential bound. $\square$

10. With model error and noisy observations

Suppose the error equation contains a disturbance $d(t)$ and observation noise $n(t)$:

$$\dot{\mathbf{e}} = (A - \mu H^\top H) \mathbf{e} + d + \mu H^\top n.$$

The same energy argument yields schematically

$$\frac{d}{dt} \|\mathbf{e}\| \leq -\lambda \|\mathbf{e}\| + \|d\| + \mu \|H^\top n\|,$$

where $\lambda = \mu\eta - \alpha > 0$.

11. Error floor under persistent disturbances

If

$$\|d\| \leq D, \quad \|H^\top n\| \leq N,$$

then comparison with a scalar ODE gives

$$\limsup_{t \rightarrow \infty} \|\mathbf{e}(t)\| \leq \frac{D + \mu N}{\lambda}.$$

Key idea

Increasing μ improves deterministic contraction, but also amplifies measurement noise. Gain tuning is therefore a bias–noise tradeoff.

12. Relation to a Luenberger observer

For linear systems, a general observer is

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + L(\mathbf{y} - H\hat{\mathbf{x}}).$$

Nudging chooses a simple gain structure, often

$$L = \mu H^\top$$

or a spatial interpolation/smoothing operator.

Interpretation

Nudging is observer theory adapted to large dynamical systems where a full optimal gain is too expensive.

13. Relation to 3D-Var

A single 3D-Var analysis minimizes

$$J(\mathbf{x}) = \frac{1}{2} \left\| \mathbf{x} - \mathbf{x}^f \right\|_{B^{-1}}^2 + \frac{1}{2} \left\| \mathbf{y} - H\mathbf{x} \right\|_{R^{-1}}^2 .$$

Its solution has a gain-like correction. Nudging can be viewed as replacing a carefully optimized covariance-weighted gain by a cheap prescribed operator γG .

14. Gain selection intuition

Too small γ :

slow correction and persistent initial-condition error.

Too large γ :

sensor noise is injected aggressively into the full state.

A practical strategy is to sweep γ on validation trajectories and choose the smallest gain that reliably synchronizes the model.

15. Minimal nudging code

```
for k in range(1, n_cycles):  
    x = nonlinear_forecast(x, dt, nu)  
    innovation = y[k] - x[obs_idx]  
    correction = periodic_interpolate(innovation)  
    x = x + gamma * correction
```

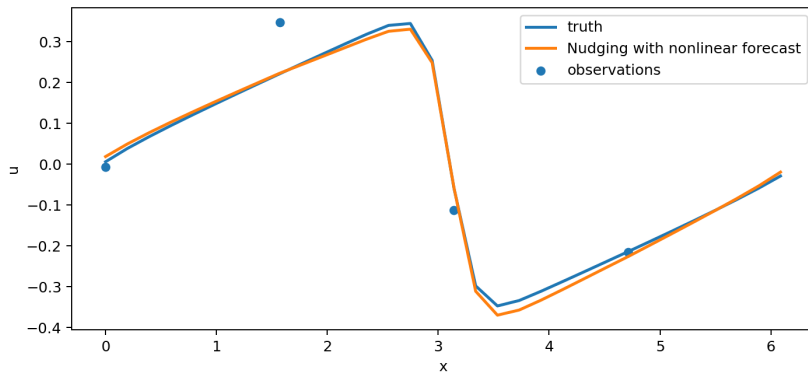
One nonlinear forecast per cycle; no ensemble and no covariance matrix.

16. Same Burgers setup, same observations

Grid	$N = 32$
Assimilation interval	0.10
Sensors	every 8th point
Observation noise	0.08
Nudging gain	$\gamma = 0.10$
Forecast model	nonlinear Burgers solver

This isolates the effect of the assimilation algorithm rather than the forecast physics.

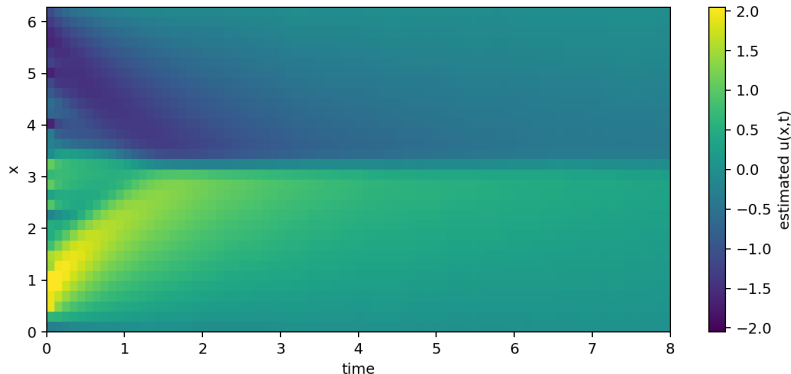
17. Nudging final-state reconstruction



Key idea

A single nonlinear trajectory plus a simple interpolated innovation reconstructs the final field closely in this controlled problem.

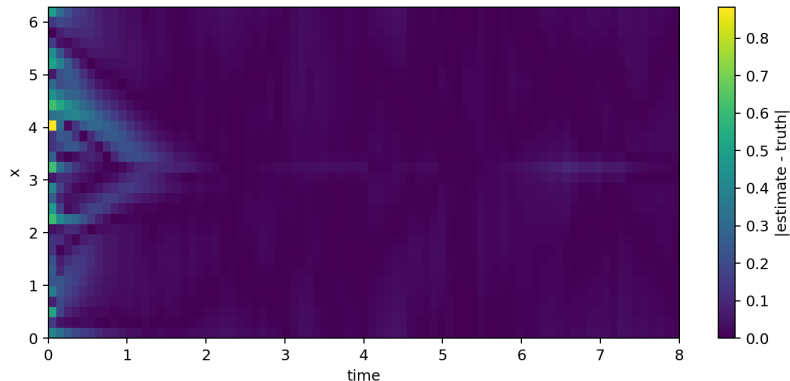
18. Nudging output over the complete trajectory



Key idea

The corrected trajectory remains synchronized with the nonlinear truth without maintaining an ensemble or covariance matrix.

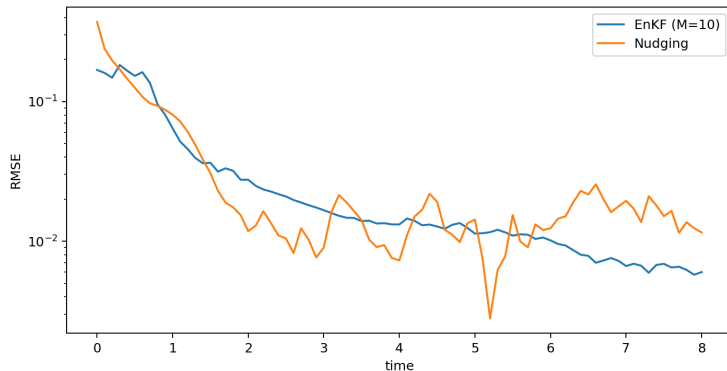
19. Nudging absolute error map



Important

The colorbar shows $|u_{\text{nudge}} - u_{\text{truth}}|$. Residual errors are concentrated and remain small compared with the KF model-mismatch map.

20. Comparable error to a small EnKF—with one forecast



Method	Mean RMSE	nonlinear forecasts/cycle
EnKF, $M = 10$	0.032	10
Nudging	0.035	1

21. Accuracy–cost tradeoff across the synthetic baselines

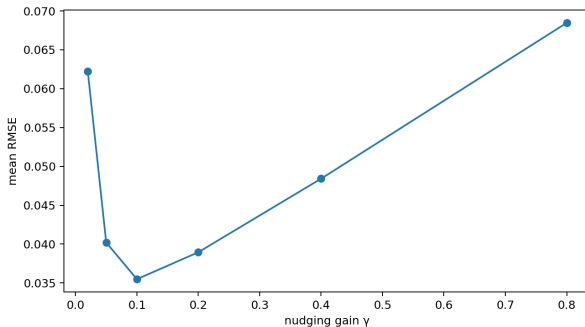
Method	Mean RMSE	nonlinear solves/cycle	Uncertainty estimate
Frozen-model KF	0.647	0*	covariance, but wrong model
EnKF, $M = 10$	0.032	10	ensemble
EnKF, $M = 40$	0.012	40	ensemble
Nudging	0.035	1	no calibrated posterior

*The KF uses one cheap linear forecast rather than the nonlinear Burgers solver.

Key idea

Nudging nearly matches the $M = 10$ EnKF error while reducing nonlinear forecast count by a factor of ten.

22. Gain sensitivity: accuracy is not free of tuning



Gain regime	Benefit	Cost / risk
Too small γ	smooth estimate	slow synchronization
Moderate γ	removes drift	best error balance here
Too large γ	fast correction	injects sensor noise

23. Why nudging can be much faster

For forecast cost $C_{\mathcal{M}}$ and inexpensive interpolation cost $C_{\mathcal{I}}$,

$$C_{\text{nudge}} \approx C_{\mathcal{M}} + C_{\mathcal{I}}.$$

An ensemble method with M members requires approximately

$$C_{\text{ensemble}} \approx MC_{\mathcal{M}} + C_{\text{statistics}}.$$

When the PDE solve dominates,

$$\frac{C_{\text{ensemble}}}{C_{\text{nudge}}} \approx M.$$

For the comparison on slide 20, this is roughly a **10:1 model-work ratio** for nearly the same mean RMSE.

24. When can nudging approach ensemble-quality state estimates?

Helpful conditions	Why they matter
Forecast model is already dynamically accurate	feedback corrects state error rather than structural model bias
Observations are frequent	unstable modes cannot drift far between corrections
Sensors constrain important spatial structures	interpolation carries useful information to unobserved points
A sensible G operator is available	innovation is spread in physically plausible directions
Primary goal is state estimation	full posterior uncertainty is not required

25. What do we give up for the speed?

Property	EnKF	Nudging
Flow-dependent gain	estimated from ensemble	prescribed / interpolated
Posterior spread	explicit ensemble	not calibrated
Model forecasts/cycle	M	1
Sampling noise	yes	no
Gain tuning	inflation/localization + ensemble size	mainly γ and G
Sparse-sensor robustness	can exploit covariance	depends strongly on G

Key idea

Nudging trades probabilistic richness and adaptive covariance for simplicity and deterministic low cost.

26. Engineering tradeoff: implementation complexity

Implementation task	EnKF	Nudging
Run many forecasts	required	no
Store many states	required	no
Compute sample covariance	required	no
Localization / inflation	often required	no
Innovation interpolation	optional	required for sparse sensors
Parallel orchestration	often important	minimal

For production CFD, this software simplicity can be as important as the asymptotic compute reduction.

27. Main takeaways

- Nudging is deterministic feedback: one nonlinear forecast followed by a prescribed innovation correction.
- The energy estimate shows how sufficiently strong observation feedback can contract state error.
- In the Burgers example, nudging reaches mean RMSE **0.035** versus **0.032** for an $M = 10$ EnKF.
- That comparable state error uses one nonlinear forecast per cycle instead of ten; a higher-accuracy $M = 40$ EnKF reaches 0.012 at still greater cost.
- The tradeoff is explicit: lower compute and simpler software in exchange for a prescribed gain and no calibrated posterior uncertainty.